Question ID: 5ae4803e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Medium

Question

$$\frac{(x+9)(x-9)}{x+9}=7$$

What is the solution to the given equation?

Answer

A. 7

B. 9

C. 16

D. 63

Correct Answer: C

Rationale

Choice C is correct. Since the left-hand side of the given equation has a factor of $x+9$ in both the numerator and the denominator, the solution to the given equation can be found by solving the equation $x-9=7$. Adding 9 to both sides of this equation yields $x=16$. Substituting 16 for x in the given equation yields $\frac{(16+9)(16-9)}{16+9}=7$, or $7=7$. Therefore, the solution to the given equation is 16.

Choice A is incorrect. Substituting 7 for x in the given equation yields $\frac{(7+9)(7-9)}{7+9}=7$, or $-2=7$, which is false.

Choice B is incorrect. Substituting 9 for x in the given equation yields $\frac{(9+9)(9-9)}{9+9}=7$, or $0=7$, which is false.

Choice D is incorrect. Substituting 63 for x in the given equation yields $\frac{(63+9)(63-9)}{63+9}=7$, or $54=7$, which is false.